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Artificial Intelligence for Differential Equations in Biomedical Engineering: From Classical Numerical Solvers to Scientific Machine Learning

Jad Dabaghi* Jinan Charafeddine *

Abstract

Differential-equation models support biomedical simulation and inference, but the value of neural methods depends on the computational task, numerical interface, and available validation evidence. This narrative methodological review compares physics-informed neural networks, neural differential equations, universal differential equations, neural operators, and hybrid frameworks with classical numerical approaches. We organize these methods by what they learn—solution fields, dynamical laws, missing mechanisms, or solution operators—and connect these targets to identifiability, physical consistency, offline–online cost, and intended use. Selected biomedical studies distinguish evidence from numerical simulations, laboratory experiments, and patient observations. Their findings support task-specific applications while showing that predictive accuracy, parameter recovery, and clinical validity require different assessments. We synthesize these comparisons into a task oriented method-selection guide, a credibility pathway, and a minimum reporting checklist. The resulting framework helps identify where learning can add value within a numerical workflow and which evidence is needed to support the resulting biomedical claims. Priorities include independent validation, uncertainty under model misspecification, and transfer across patients, anatomies, and interventions.

Keywords: Biomedical engineering, differential equations, numerical analysis, scientific machine learning, physics-informed neural networks, neural operators, digital twins.

1 Introduction

Mathematical models connect physiological mechanisms to predictions: ordinary differential equations (ODEs) describe temporal dynamics, whereas partial differential equations (PDEs) resolve spatial transport, deformation, and propagation [51, 71]. Cardiac models couple cellular activity, tissue conduction, and sometimes mechanics [94, 99]. On realistic anatomies, numerical discretization and time integration generally replace unavailable closed-form solutions [9, 31, 32, 46, 83]. Error estimation supports numerical verification, while independent measurements are needed for physiological validation [82, 103]. Repeated solves for calibration, uncertainty analysis, and virtual cohorts motivate acceleration [72]; digital twins also require individualized updating [18]. Reduced-order models (ROMs) and parallel-in-time methods provide relevant numerical alternatives to neural acceleration [22, 35, 36, 43, 59, 68, 79].

Scientific machine learning (SciML) integrates trainable representations with equations, numerical algorithms, and scientific constraints [20, 50, 108]. Physics-informed neural networks (PINNs) fit observations and differential-equation residuals within an optimization problem [81]. Neural ODEs parameterize continuous-time dynamics evaluated by numerical integration [15]. Universal differential equations (UDEs) introduce trainable components into mechanistic systems [80]. DeepONet and Fourier neural operators (FNOs) learn mappings between input functions and solution fields across families of problems [58, 66]. These approaches address different computational tasks and should not be ranked as interchangeable solvers.

Pressure inference from flow MRI, aortic-damage surrogacy, and cross-regimen drug forecasting illustrate distinct inverse, operator, and temporal-generalization tasks [53, 65, 111]. Their evaluation requirements differ. Existing reviews establish broad SciML foundations, and Ahmadi et al. provide the closest biomedical methodological context [3]. The contribution of this narrative review is a joint analysis of classical numerical methods and neural approaches:

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Table 1: Positioning of the present review relative to closely related reviews. The comparison concerns primary emphasis and the organization of the present synthesis; it does not imply that individual topics are absent from previous reviews.

Review	Primary emphasis	Relationship to the present synthesis
Karniadakis et al. [50]	Broad foundations and perspectives on physics-informed machine learning.	Provides the general methodological context. The present review organizes biomedical applications around computational tasks, numerical interfaces, and validation requirements.
Cuomo et al. [20]	Scientific machine learning through PINNs, including methodology and research directions.	Provides a PINN-focused methodological foundation. The present review compares residual fitting with learned dynamics, operator approximation, and solver-based hybrid approaches.
Willard et al. [108]	Integration of physics-based modelling with machine learning.	Provides a broad integration framework. The present review applies a task-oriented assessment to biomedical differential equations, linking cost, identifiability, and evidence provenance.
Ahmadi et al. [3]	PINNs, neural ODEs, and neural operators in biomedical science and engineering.	Provides the closest biomedical methodological context. The present review foregrounds the joint assessment of numerical-solver interfaces, identifiable inference, offline–online cost, and validation scope.
Present review	**Task-oriented comparison of numerical and learning-based biomedical models.**	**Connects five assessment axes to a method-selection guide, a credibility pathway, and a minimum reporting checklist.**

it identifies where learning enters a numerical pipeline, what computation remains at deployment, and which evidence supports each biomedical claim. The synthesis connects five assessment axes—solver interface, identifiable biological inference, offline–online cost, validation provenance, and task-oriented selection—to a method-selection guide, a credibility pathway, and a minimum reporting checklist. Table 1 clarifies this positioning without claiming a new numerical or learning algorithm.

Section 2 defines the review scope. Section 3 introduces governing models, numerical discretization, and acceleration; Section 4 explains how neural methods approximate solutions, dynamics, or operators and interface with solvers. Section 5 examines biomedical applications. Sections 6 and 7 connect comparative evidence to selection and reporting criteria, and Sections 8 and 9 address translation and research priorities.

2 Review Approach and Scope

2.1 Narrative Design and Retrieval

This narrative methodological review combines numerical and physiological foundations with purposively selected original SciML studies [107]. It is not a systematic review or meta-analysis conducted under PRISMA 2020 [74]: it does not claim exhaustive retrieval, pooled effect estimates, or formal study-quality scores. Coverage emphasizes developments since 2018 and incorporates targeted updates from 2024 to 2026. Targeted queries paired PINN, neural differential equation, UDE, DeepONet, FNO, and hybrid-learning terms with hemodynamics, electrophysiology, biomechanics, perfusion MRI, pharmacokinetics, glioblastoma, brain transport, migration, and respiratory flow. The author-reported checks on 9 October 2026 used publisher pages, PubMed-indexed research, institutional records, proceedings, and identified preprints. These checks were not a database-wide systematic search. Foundational sources also came from the existing bibliography.

Selection sought explicit governing models, learned roles, biomedical outputs, and contrasting evaluation sources: simulations, laboratory measurements, and patient observations. Reviews provide context rather than original application evidence. Purposive selection may favor well-documented studies and cannot establish method prevalence. Quantitative details unavailable in inspected sources were not reconstructed.

Tables 4 and 5 record task, provenance, findings, and evidence limits. Results remain study-specific; no new experiments or comparative benchmark runs were performed. The selection guide, checklist, and credibility pathway are proposed syntheses, not validated decision rules.

2.2 Scope and Boundaries

The focus is PDE-based biomedical modeling, with ODEs included for continuous-time learning, compartmental physiology, and spatial coupling. Tasks include forward solution, inverse inference, constitutive learning, equation

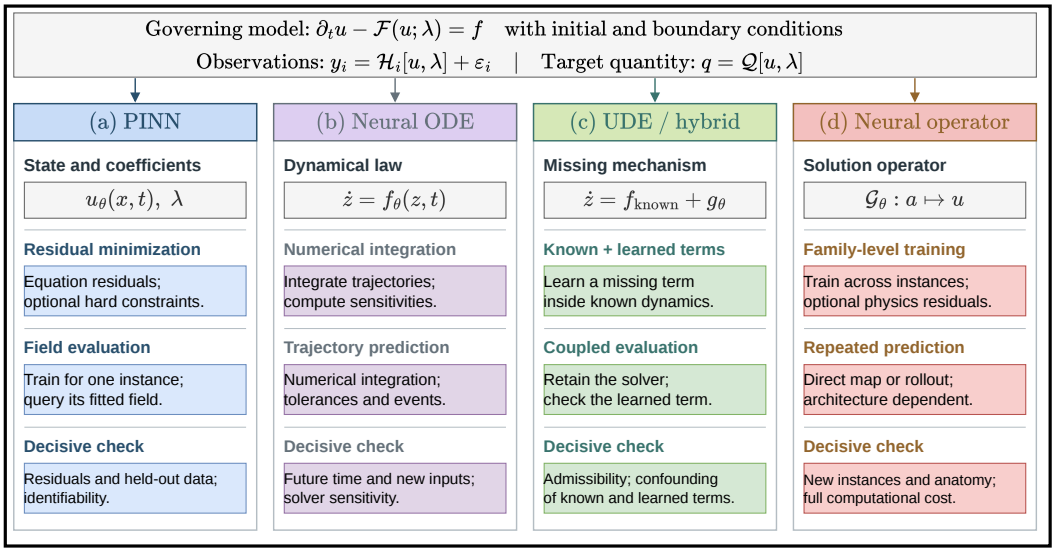


Figure 1: Computational roles of neural methods for differential equations: (a) PINNs approximate states and, in inverse problems, coefficients; (b) neural ODEs learn dynamical laws; (c) UDEs combine known dynamics with learned terms; and (d) neural operators learn mappings from input functions to solution fields. Each column summarizes the learned object, training or evaluation strategy, and key checks. Here, z denotes the state vector and a an input function. These categories are not mutually exclusive.

discovery, and repeated-query prediction. Pure classification and unconstrained forecasting are excluded unless linked to differential equations. Application coverage is uneven; tissue engineering, rehabilitation, prostheses, and integrated digital twins remain opportunities where selected evidence does not establish implementation. Figure 1 summarizes the taxonomy.

3 Mathematical and Numerical Foundations

3.1 Governing Models and Observation Operators

Let $\Omega \subset \mathbb{R}^d$, $d = 2, 3$, be a bounded connected domain with Lipschitz boundary, and let $T > 0$ denote the final simulation time. We set $\Omega_T = \Omega \times (0, T)$. Polygonal or polyhedral computational domains are common approximations to anatomical geometries. A general evolution problem is

$$\begin{aligned} \partial_t u - \mathcal{F}(u;\lambda) &= f && \text{in } \Omega_T, \\ \mathcal{B}(u;\lambda) &= g && \text{on } \partial\Omega \times (0,T), \\ u(\cdot,0) &= u_0 && \text{in } \Omega. \end{aligned} \tag{1}$$

Here u is a scalar field or a vector of physical fields, λ contains physical parameters or coefficient fields, and $\mathcal{F}$ denotes a spatial differential operator that may depend on u, ∇u, and higher spatial derivatives. The source f, boundary operator $\mathcal{B}$, prescribed boundary data g, and initial state u_0 complete the problem. For example, $\mathcal{F}(u;\kappa) = \nabla \cdot (\kappa \nabla u)$ gives a diffusion equation; for constant scalar $\kappa > 0$, this is the heat equation $\partial_t u - \kappa \Delta u = f$. The notation is an organizing template, not a claim that every biomedical system is parabolic: incompressible flow also includes a pressure constraint, and quasistatic mechanics omits the temporal derivative [78].

Measurements generally observe only part of the state:

$$y_i = \mathcal{H}_i[u,\lambda] + \varepsilon_i. \tag{2}$$

The observation operator $\mathcal{H}_i$ may include spatial averaging or an indirect imaging-to-state relationship. It must be specified before interpreting image-derived parameters [101].

ODEs represent lumped temporal states. Fractional derivatives introduce nonlocal memory [76], whereas stochastic differential equations describe process variability [44]; neither should be confused with stochastic optimization.

3.2 Representative Biomedical Equations

For an incompressible Newtonian approximation to blood or airflow,

$$\rho(\partial_t \mathbf{v} + \mathbf{v} \cdot \nabla \mathbf{v}) = -\nabla p + \mu \nabla^2 \mathbf{v} + \mathbf{f}, \tag{3}$$

$$\nabla \cdot \mathbf{v} = 0. \tag{4}$$

Applicability depends on scale, geometry, inflow, outlet impedance, and wall behavior [18].

The diffusion example extends to cardiac electrophysiology by adding ionic reaction dynamics and anisotropic tissue conductivity. A monodomain-type formulation is

$$\begin{aligned} \chi C_m \partial_t V &= \nabla \cdot (\mathbf{D} \nabla V) - \chi I_{\text{ion}}(V, \mathbf{w}) + I_{\text{app}}, \\ \partial_t \mathbf{w} &= \mathbf{q}(V, \mathbf{w}), \end{aligned} \tag{5}$$

An insulating boundary condition $\mathbf{D}\nabla V \cdot \mathbf{n} = 0$ and initial states $V_0, \mathbf{w}_0$ complete the model. Here V is transmembrane potential, $\mathbf{w}$ collects ionic states with kinetics $\mathbf{q}$, $\mathbf{D}$ is effective anisotropic conductivity, χ is membrane surface area per tissue volume, and C_m is capacitance per membrane area. The ionic current I_{ion} is specified per membrane area and I_{app} per tissue volume. Diffusion coupled to ionic reactions connects this system to (1). Bidomain models retain separate intra- and extracellular potentials; eikonal models describe activation times without the full potential dynamics [16, 94].

A generic tissue transport or growth model is

$$\partial_t c + \nabla \cdot (\mathbf{v} c) = \nabla \cdot (\mathbf{D} \nabla c) + r(c, \lambda). \tag{6}$$

Transport, production, and proliferation share this structure, but require context-specific parameters and observations [71]. Quasistatic solid-mechanics balance is

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{b} = 0, \qquad \boldsymbol{\sigma} = \mathcal{C}(\boldsymbol{\epsilon}, \lambda), \tag{7}$$

Here $\boldsymbol{\sigma}$ is Cauchy stress, $\mathbf{b}$ is body force per current volume, and $\boldsymbol{\epsilon}$ is small strain. The constitutive map $\mathcal{C}$ determines material behavior [45]; finite-strain extensions require consistent configurations and stress measures.

3.3 Classical Discretization Methods

Finite differences approximate nodal derivatives, usually on structured grids [57]. Finite elements use weak formulations and mesh-based basis functions for complex geometries [9, 46]. Finite volumes enforce control-volume balances through conservative interface fluxes [31, 32]. DG methods couple independent elementwise polynomial traces using numerical fluxes and stabilization [27, 83]; hybrid high-order methods couple cell and face unknowns on general meshes [28]. Table 2 summarizes their numerical and learning interfaces.

3.4 Acceleration of Numerical Methods

Projection-based ROMs construct representative solutions and a reduced basis offline, then solve a smaller system online; nonlinear terms may require hyper-reduction [43, 68, 79]. Preserving physical structure requires specific construction [21]. Parallel-in-time methods distribute work over temporal subintervals [14, 73]. Parareal combines coarse propagation, parallel fine solves, and iterative correction; efficiency depends on convergence, processor availability, and communication [22, 35, 36, 59]. Both approaches provide task-dependent baselines for neural acceleration.

3.5 From Numerical Discretization to Learning Procedures

For the diffusion example, a conforming finite-element approximation is $u_h(\mathbf{x}, t) = \sum_{a=1}^{N_h} U_a(t)\phi_a(\mathbf{x})$. After boundary treatment, spatial discretization gives

$$\mathbf{M}\dot{\mathbf{U}}(t) + \mathbf{A}(\kappa)\mathbf{U}(t) = \mathbf{F}(t), \tag{8}$$

Here $\mathbf{U}(t) = [U_1(t), \ldots, U_{N_h}(t)]^\top$ contains time-dependent coefficients, $\mathbf{M}$ is the mass matrix, $\mathbf{A}$ the diffusion stiffness matrix, and $\mathbf{F}$ the load. For nodal Lagrange elements, $U_a(t) \approx u(\mathbf{x}_a, t)$; other bases need not have this interpretation. Time integration approximates $\mathbf{U}$ at discrete instants. Learning can target the solution, missing dynamical terms, or an input-to-solution map.

Table 2: Numerical methods and their relationship to SciML. Advantages are conditional on suitable formulation and implementation.

Method	Numerical role	Relevant strengths and costs	Possible learning interface
Finite differences [57]	Nodal derivative approximation	Efficient on structured grids; boundary and stability treatment matter	Learned stencils or correction terms
Finite elements [46]	Weak-form spatial approximation	Geometric flexibility; mesh, assembly, and nonlinear-solve costs	Material law, reduced basis, solver differentiation
Finite volumes [32]	Integral conservation	Explicit flux balance; reconstruction and near-wall resolution matter	Learned flux or closure with balance constraints
DG / high-order methods [27]	Local polynomial spaces and interface coupling	Local refinement; flux, penalty, and degree choices	Local operators and weak-residual training
Reduced-order models [79]	Projection onto a small state space	Fast repeated evaluation after offline construction	Learned latent dynamics or operator surrogate

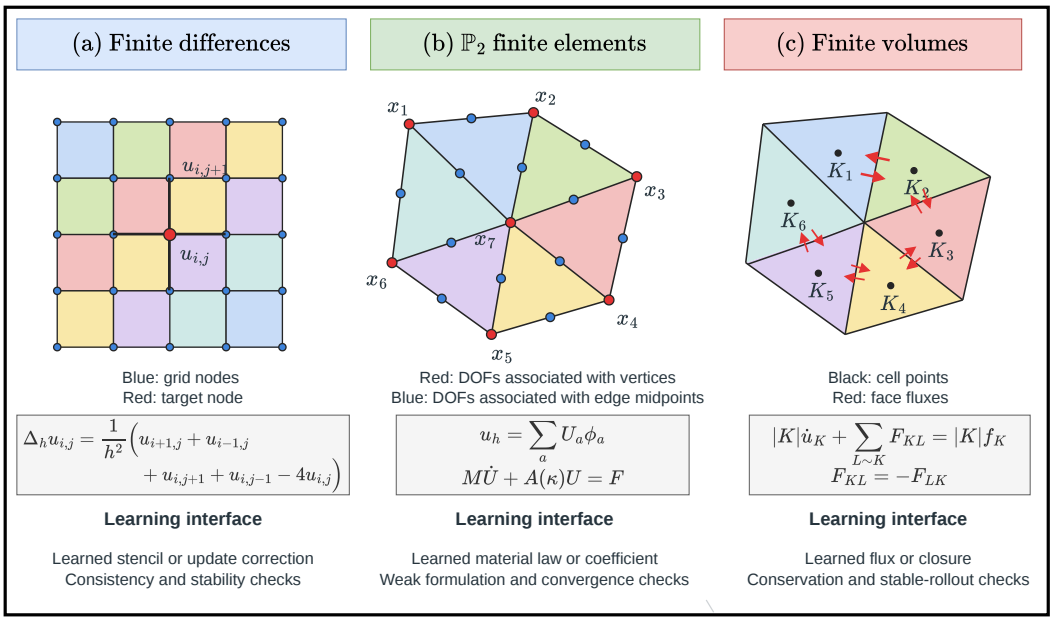


Figure 2: Comparison of discrete representations and potential learning interfaces. (a) Finite differences: nodal values on a uniform Cartesian grid and a five-point Laplacian stencil. (b) Quadratic Lagrange finite elements ($\mathbb{P}_2$): degrees of freedom at triangle vertices (red) and edge midpoints (blue). (c) Cell-centered finite volumes: cell unknowns and oriented interface fluxes satisfying $F_{KL} = -F_{LK}$. These representations provide interfaces for learned stencil corrections, constitutive relations, and flux closures, respectively. Cell colors distinguish mesh elements and have no physical meaning.

Figure 2 illustrates learned stencil corrections, constitutive responses, and conservative fluxes. Learned discretizations provide one direct interface [6]. Equation (2) defines the observation link, and Section 7 specifies evaluation and reporting.

3.6 Error, Stability, and Adaptivity

A priori estimates relate approximation error to discretization and solution regularity; Céa's lemma illustrates the coercive Galerkin setting [13]. A posteriori estimators use computed solutions and problem data to guide refinement; guaranteed bounds and separation of spatial, temporal, or algebraic errors require stated assumptions [23, 24, 37, 82,

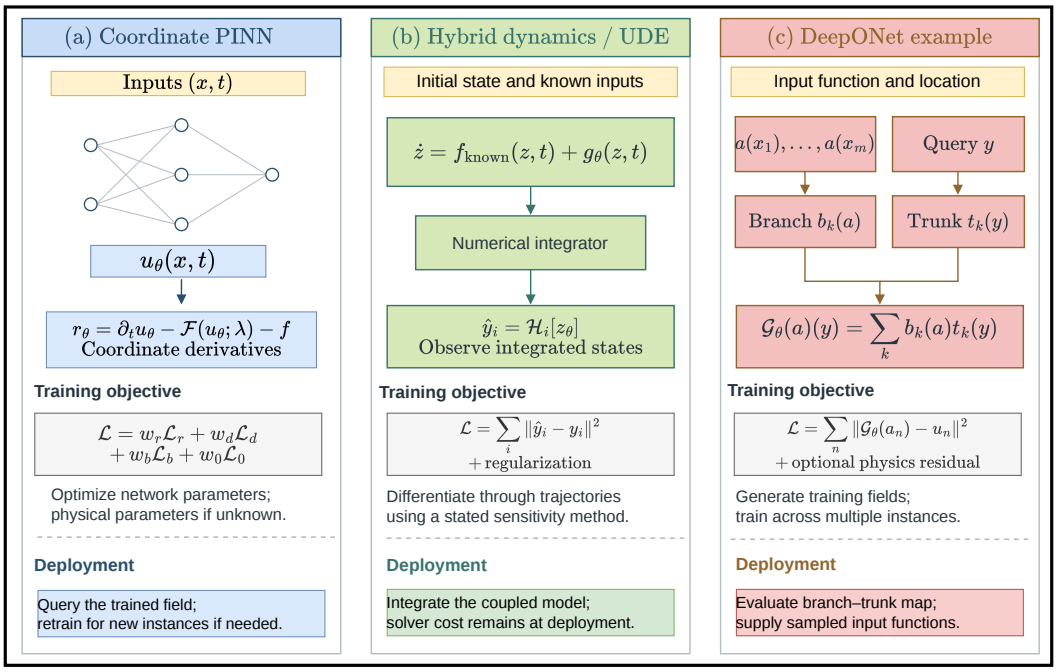


Figure 3: Training and deployment pathways for (a) coordinate PINNs, using equation residuals and data or initial/boundary constraints; (b) UDEs, combining known and learned dynamics within a numerical integrator; and (c) DeepONet, combining branch and trunk networks to predict solution fields. The diagrams highlight the distinct training objectives and computational steps retained at deployment.

103]. Learning additionally introduces representation, sampling, and optimization errors. Independent residual and reference-solution checks, and reassessment of modified fluxes or constitutive laws, remain necessary.

4 Artificial Intelligence Methods for Differential Equations

4.1 Computational Roles and Numerical Interfaces

Neural methods for differential equations differ in their learned target and in how they interact with numerical computation. A network may approximate the solution u of a problem such as (1), unknown parameters λ, a dynamical law, or a solution operator $\mathcal{G} : a \mapsto u$. PINNs typically approximate states through residual-based optimization; neural ODEs learn dynamics evaluated by numerical integration; UDEs augment known mechanisms with learned terms; and neural operators approximate solution maps across problem instances. These choices determine the required training information and the computation retained at deployment.

Physics can enter the training objective through residual penalties, the architecture through constraints, or a numerical solver through enforced balance laws [108]. These categories overlap: a UDE can contain PDE dynamics, a neural operator can use residual losses, and a PINN can infer constitutive terms. Consequently, the method label alone does not establish conservation, computational advantage, or reliable generalization. Figure 3 illustrates the training and deployment pathways.

4.2 Physics-Informed Neural Networks

4.2.1 Residual Formulation

PINNs represent the solution by a network $u_\theta(\mathbf{x}, t)$ and penalize departures from the governing equations [81]. For (1), the strong residual is

$$r_{\theta,\lambda} = \partial_t u_\theta - \mathcal{F}(u_\theta; \lambda) - f. \tag{9}$$

Weak or variational formulations provide an alternative: the PDE is tested against suitable functions and integrated by parts, reducing spatial derivative requirements. Their implementation requires appropriate test spaces, quadrature, and boundary treatment [20].

A typical objective is

$$\mathcal{L} = w_d\mathcal{L}_d + w_r\mathcal{L}_r + w_b\mathcal{L}_b + w_0\mathcal{L}_0. \tag{10}$$

The nonnegative weights w_d, w_r, w_b, w_0 balance four discrepancies: $\mathcal{L}_d$ measures mismatch with observed data, $\mathcal{L}_r$ measures the interior PDE residual, $\mathcal{L}_b$ measures boundary-condition violations, and $\mathcal{L}_0$ measures initial-condition violations. For sample sets of sizes N_d, N_r, N_b, N_0, a representative strong-form construction is

$$\begin{aligned}
\mathcal{L}_d &= \frac{1}{N_d}\sum_{i=1}^{N_d} \|\mathcal{H}_i[u_\theta, \lambda] - y_i\|^2, \\
\mathcal{L}_r &= \frac{1}{N_r}\sum_{j=1}^{N_r} |r_{\theta,\lambda}(\mathbf{x}_j^r, t_j^r)|^2, \\
\mathcal{L}_b &= \frac{1}{N_b}\sum_{j=1}^{N_b} |\mathcal{B}(u_\theta; \lambda)(\mathbf{x}_j^b, t_j^b) - g(\mathbf{x}_j^b, t_j^b)|^2, \\
\mathcal{L}_0 &= \frac{1}{N_0}\sum_{j=1}^{N_0} |u_\theta(\mathbf{x}_j^0, 0) - u_0(\mathbf{x}_j^0)|^2.
\end{aligned} \tag{11}$$

Vector-valued equations use compatible norms, and noise-aware data terms may use observation covariance weighting. Minimization adjusts network parameters θ; an inverse problem additionally optimizes selected physical parameters λ. Automatic differentiation evaluates derivatives of the network representation, but does not remove representation error or uncertainty in the governing model [81].

Residual norms, collocation, nondimensionalization, and noise-weighted data terms affect optimization; correlated image pixels need not identify an unknown coefficient.

4.2.2 Constraints and Training Difficulty

Penalty enforcement is approximate. Dirichlet conditions can sometimes be built into the representation, for example $u_\theta = g + \psi N_\theta$ with $\psi = 0$ on the prescribed boundary. This construction must be compatible with the geometry and regularity of the desired solution.

Domain decomposition and adaptive sampling address the distribution and scale of training [20]. Composite losses can yield unbalanced gradients, and optimization may struggle even for relatively simple equations [106]. Failure-mode studies show that increasing differential-equation complexity can degrade a standard PINN despite apparently adequate expressive capacity [55]. These observations motivate balanced losses, curriculum and adaptive sampling strategies, and problem-specific training, rather than an assumption that a physics term automatically solves data scarcity.

Inverse identifiability requires distinguishing coefficients from uncertain boundary forcing, observation error, and omitted physiology. Fitting a field alone does not establish a biological parameter.

4.2.3 Uncertainty and Appropriate Use

Bayesian PINNs combine priors and noisy data with a physics-informed likelihood or penalty structure, providing a route to posterior inference [110]. The resulting uncertainty is conditional on the assumed model and inference approximation. An ensemble of trained networks may characterize optimization variability or predictive disagreement, but does not automatically encompass model discrepancy.

PINNs suit sparse-data reconstruction and inverse estimation. Per-instance training and continuous field queries should be compared with the corresponding numerical inverse baseline, including training cost.

4.3 Neural Differential Equations

4.3.1 Neural ODEs and Numerical Integration

A neural ODE specifies

$$\begin{aligned} \dot{\mathbf{z}}(t) &= f_\theta(\mathbf{z}(t), t), \\ \mathbf{z}(t_1) &= \mathbf{z}(t_0) + \int_{t_0}^{t_1} f_\theta(\mathbf{z}(s), s)\,\mathrm{d}s. \end{aligned} \tag{12}$$

The network f_θ defines a vector field governing the time derivative of the state vector $\mathbf{z}(t) \in \mathbb{R}^N$; a numerical integrator evaluates the trajectory. The components $z_j(t)$ are scalar functions of time. If the ODE arises from a nodal PDE discretization, one may choose $z_j(t) \approx u(\mathbf{x}_j, t)$, as in (8); a general neural ODE may instead evolve compartmental variables or latent features, without any spatial-node interpretation. Discrete observation times $t_0 < \cdots < t_n$ specify requested states, while an adaptive solver may take different internal steps [15].

A continuous-time latent state can be decoded at requested observation times, but it need not correspond to a physiologically interpretable state. Smoothness of the representation alone does not establish correct physiology.

Explicit adaptive Runge–Kutta methods are useful for nonstiff dynamics [30, 40]. Stiff learned dynamics may require implicit integration, appropriate tolerances, and additional Jacobian evaluations [41]. The solver is therefore part of the model specification. Training with one tolerance and evaluating with another can change the effective trajectory and prediction error.

Discrete differentiation and continuous adjoints trade memory against additional solves and sensitivity error. Report integrator, tolerances, event treatment, and gradient strategy.

4.3.2 Latent and Controlled Dynamics

Latent ODEs accommodate irregular observations through probabilistic latent dynamics [85]; inference and observation models must account for missingness and the intended clinical state.

Neural controlled differential equations evolve a latent state in response to a continuous representation of the observed input path [52]. Extensions address online prediction from irregular time series [70]. Interpolation rules and treatment of observation times become part of the computational assumptions. Such methods are related to differential-equation learning, but they should not be described as enforcing a known PDE unless that constraint is actually present.

A bolus dose is a drug amount administered over a short interval. Such doses and other interventions may require explicit inputs, event handling, and state resets consistent with the relevant mass balance.

4.4 Universal Differential Equations

A UDE separates an established mechanism from an unknown component:

$$\dot{\mathbf{x}} = f_{\mathrm{known}}(\mathbf{x}, t; \lambda) + g_\theta(\mathbf{x}, t). \tag{13}$$

Here f_{known} represents established mechanisms, g_θ learns an unresolved contribution, and λ contains physical parameters such as reaction or exchange rates. Training compares integrated observations with measurements and may impose structural constraints. The UDE viewpoint also accommodates spatial models [80].

A learned term may absorb incorrect physics, missing inputs, or observation bias; joint physical–neural identifiability therefore requires independent assessment. A recent preprint distinguishes parametric identifiability of mechanistic components from functional identifiability of learned terms in chemical-reaction-network UDEs [62]. Equal-and-opposite exchange fluxes, positive rate parameterizations, or admissible constitutive energies can enforce selected properties, but conservation is not automatic. Compare mechanistic, data-only, and hybrid baselines, counting integration and sensitivity costs. Schmid et al. compared ensemble, variational, and Markov-chain Monte Carlo uncertainty approaches in three synthetic UDE examples [91]; this is simulation evidence rather than biological validation.

4.5 Neural PDEs and Equation Discovery

Networks can learn missing spatial dynamics or differential operators:

$$\partial_t u = \mathcal{F}_{\text{known}}(u; \lambda) + \mathcal{N}_\theta(u, \nabla u, \mathbf{x}, t). \tag{14}$$

The known term includes prescribed forcing. Training can differentiate through a discretized solver or minimize equation residuals; these implementations incur different numerical errors and costs. PDE-Net learns constrained convolutional differential approximations, and PDE-Net 2.0 adds numerical–symbolic structure [63, 64].

Sparse discovery instead selects evolution terms from a candidate library [11, 86]. Graph-structured oscillator extensions and biological oscillator studies illustrate its scope and limitations [7, 77]. Wu et al. combine partially known dynamics, neural missing terms, and sparse regression [109]. Cell-migration assays provide a biomedical example of learning diffusion and growth responses [56]. Noise, derivative estimation, unobserved states, and library choice can make the inferred mechanism ambiguous; independent conditions or interventions should test its interpretation.

4.6 Neural Operators

4.6.1 Learning a Family of Solutions

A neural operator approximates a solution map $\mathcal{G} : a \mapsto u$ over a family of coefficients, sources, or initial/boundary conditions. For the heat equation with other inputs fixed, $a = \kappa(\mathbf{x})$ may represent scalar or tensor-valued diffusion. Numerical solvers provide supervised pairs (a_i, u_i); after training, $\mathcal{G}_\theta(a)$ predicts solutions without a reference solve for each direct query. Accuracy on unseen inputs, resolutions, or anatomies requires separate evaluation.

A representative neural-operator layer combines local transformations and nonlocal interactions [54]:

$$v_{\ell+1}(\mathbf{x}) = \sigma\left(W_\ell v_\ell(\mathbf{x}) + \int_\Omega k_\ell(\mathbf{x}, \mathbf{y}) v_\ell(\mathbf{y})\, \mathrm{d}\mathbf{y} \right). \tag{15}$$

Here v_ℓ is a feature field, W_ℓ a pointwise linear map, and k_ℓ a learned spatial kernel. Lifting and projection connect features to physical inputs and outputs; some architectures also condition the kernel on the input. Samples and quadrature implement the integral, so shared parameters alone do not establish accuracy across discretizations.

4.6.2 DeepONet

DeepONet uses a branch network to encode sampled input functions and a trunk network to encode output coordinates [66]. A common representation is

$$\mathcal{G}_\theta(a)(y) = \sum_{k=1}^{p} b_k(a(x_1), \ldots, a(x_m)) t_k(y) + b_0. \tag{16}$$

This form separates the problem instance from the evaluation location. The choice and number of input sensors determine which function information can be supplied. Continuous output coordinates do not make the input sampling irrelevant. Training data, geometry encoding, and the distribution of physical parameters still define the supported operating range.

DeepONet is useful for repeated parameter-to-field prediction and inverse estimation using a trained surrogate. Aortic dissection provides an application in which phase-field finite-element simulations generate operator-learning data [111]. That connection makes the numerical solver a source of supervisory evidence, rather than a component rendered unnecessary by the network.

4.6.3 Fourier Neural Operators

An FNO parameterizes part of the integral operation in Fourier space:

$$v_{\ell+1} = \sigma\left(W_\ell v_\ell + \mathcal{F}^{-1}[R_\ell \mathcal{F}(v_\ell)] \right). \tag{17}$$

In (17), $\mathcal{F}$ denotes the Fourier transform and $\mathcal{F}^{-1}$ its inverse, rather than the differential operator in (1); R_ℓ contains learned spectral weights.

The learned transform acts on retained modes [58]. Spectral structure can make evaluation efficient on regular grids. Nonperiodic boundaries, irregular anatomy, and sharp interfaces require appropriate representation and evaluation rather than reliance on the generic architecture alone.

For biomedical fields, assess local wall gradients and boundary behavior alongside bulk error, including truncation and resampling effects.

4.6.4 Graph Operators and Geometry

Graph-based operator representations replace a fixed Cartesian organization with interactions among sampled points or mesh nodes. Coordinates, connectivity, and edge features can encode spatial relationships [54]. This is appealing for anatomical domains, but a graph neural network is not automatically a discretization-consistent operator. Neighborhood construction, quadrature weights, and geometry variation must be documented.

The G-FuNK preprint combines a domain-adapted graph Laplacian with learned spectral components and numerical ODE integration [61]. Its numerical experiments include diffusion, reaction–diffusion, and cardiac electrophysiology across varying domains and anisotropic diffusivity fields, illustrating an operator–dynamics hybrid rather than a purely direct temporal predictor.

Hold out complete anatomies when claiming patient transfer; mesh-resolution, shape, topology, and boundary shifts test different capabilities.

4.6.5 Physics-Informed Operators and Limitations

Residual penalties can supplement supervised operator learning, allowing data and equation information to constrain a problem family. They add derivative or solver costs during training and retain the limitations of the governing model. The operator's offline cost, the expense of generating simulation data, and retraining under distribution shift should be included when claiming acceleration.

DeepONet exposes coordinates, FNO exploits spectral grid structure, and graph operators encode connectivity. None alone guarantees conservation or transfer; Table 3 summarizes their computational roles.

4.7 Hybrid Physics–AI Frameworks

4.7.1 Learned Components Within Solvers

Hybrid solvers learn fluxes, constitutive laws, unresolved terms, preconditioners, or coarse-model corrections. Learned discretizations illustrate this interface [6]. Assess discrete conservation and long-horizon stability in the coupled solver. Zhao et al. proposed a bifurcation-based domain decomposition coupling neural operators with Schwarz waveform relaxation for one-dimensional blood-flow simulation [114]. Constitutive learning from biaxial aortic measurements provides a biomedical example [60]; its evidence concerns the tested tissue response, not every downstream organ simulation.

4.7.2 Multi-Fidelity and Reduced Representations

Multi-fidelity models combine cheaper simulations with higher-fidelity or measured data; the learned discrepancy may concern numerical error, closures, parameters, or observations. Reduced latent dynamics can couple a classical basis to learned evolution. Compare with structure-preserving ROMs and assess projection, temporal, and output-reconstruction errors [21].

4.7.3 Digital-Twin Integration

A surrogate is a candidate twin component until patient linkage, calibration, and an updating mechanism are demonstrated [18, 104]. Where latency permits, a numerical reference can support fallback or periodic verification.

Table 3 separates learned targets, equation interfaces, and deployment costs. Operator architecture alone does not enforce physics; solver-generated data and explicit constraints provide different forms of equation information.

Table 3: Comparison of SciML paradigms by computational role and equation interface. Entries describe common formulations rather than universal performance guarantees.

Paradigm	Learned target	Equation interface	Computational cost	Key assessment issues
PINN [81]	Solution fields; unknown parameters in inverse problems	Equation residuals and initial/boundary constraints	Typically per-instance training, followed by field queries	Loss balance, independent residuals, and identifiability
Neural ODE [15]	State or latent-state dynamics	Learned vector field evaluated by numerical integration	Integration during training and inference; gradient computation during training	Solver tolerances, stiffness, events, and predictive generalization
UDE [80]	Unknown component of a mechanistic model	Learned terms embedded in governing dynamics	Numerical solves during training and inference; sensitivities during training	Physical–neural confounding, identifiability, and admissibility
Neural PDE [63]	Spatial dynamics or equation terms	Learned differential operators or terms within a discretized evolution model	Training and rollout costs depend on resolution and prediction horizon	Discretization consistency, noise sensitivity, and rollout stability
DeepONet [66]	Function-to-field map	Supervised solution data; optional equation residuals	Offline data generation and training; direct field queries after training	Input sampling, instance distribution, and surrogate error
FNO / graph [54]	Solution operator over a problem family	Supervised solution data; optional equation residuals	Offline data generation and training; operator evaluation, with integration for some temporal architectures	Geometry, boundaries, resolution, and distribution shift
Hybrid solver [6]	Flux, closure, or numerical correction	Learned component embedded in a numerical solver	Numerical solution plus neural evaluation; offline training	Discrete conservation, consistency, and long-horizon stability

5 Biomedical Differential-Equation Landscape

5.1 Cardiovascular Systems and Hemodynamics

Hemodynamic learning targets velocity, pressure, boundary conditions, or quantities derived from these fields. Kissas et al. combined a reduced arterial-flow model with noninvasive 4D flow MRI to infer pressure [53]. This application demonstrates data–equation integration for an inverse problem; its validity remains conditional on the arterial model and measurement interpretation. Sen et al. combined one-dimensional blood-flow equations with physics-informed graph neural networks, evaluating predictions against numerical references and in vivo murine measurements [93].

Recent work extends MRI-based hemodynamic inference to pulmonary arterial pressure using a reduced-order Navier–Stokes model [47]. Villié et al. investigate artifact correction and assimilation of mean pressure and Reynolds stresses from in vitro stenosis flow-MRI measurements [105]. Aghaee and Khan examine Fourier-based activation functions within PINNs for patient-specific cardiovascular flows [2]. Their targets are inverse estimation, measurement assimilation, and representation, respectively.

Cai et al. reconstructed three-dimensional flow information using artificial-intelligence velocimetry and a microaneurysm-on-a-chip setting [12]. Arzani et al. investigated near-wall flow inference from sparse information [5]. These studies address distinct observation scales. Experimental microfluidic evidence should not be relabeled as a clinical arterial cohort, and near-wall reconstruction deserves assessment separately from whole-domain velocity error.

Pressure inference requires consistent pressure-reference conventions and boundary conditions, while wall shear stress is particularly sensitive to near-wall velocity gradients and geometry. Agreement of velocity contours alone therefore does not establish the accuracy of these derived quantities. Evaluation should use synchronized measurements and calibrated numerical inverse methods where available.

Two recent studies illustrate different evidence settings. Velikorodny et al. combined DeepONet with multifidelity training for stenosed-vessel flow. The authors reported experimental–model differences in instantaneous pressure

of 1.2 ± 3.5 mmHg for the multifidelity model [102]. This is controlled in vitro agreement, not patient-outcome validation. Ryan et al. embedded a physics-informed operator for a 17-vessel network in Bayesian inference, using sparse flow MRI in two Fontan patients and comparison with cuff pressures [87]. The small clinical example supports feasibility; it does not establish population transportability or unique recovery of all vascular parameters.

Csala et al. coupled learned blood-flow dynamics to physical constraints and compared predictions with three-dimensional simulation references and a one-dimensional FEM baseline [19]. The example strengthens the case for improving reduced numerical models through learned terms, while retaining a simulation-only evidence boundary. Pegolotti et al. develop a graph-neural-network reduced-order model for cardiovascular simulations [75]. This is learned reduced dynamics rather than residual-based inverse inference.

5.2 Cardiac Electrophysiology

Sahli Costabal et al. used an eikonal-constrained network to reconstruct activation time and conduction velocity from sparse samples and studied uncertainty-guided sampling [88]. The evidence in that study is simulation based. Eikonal constraints provide a reduced description of propagation; they should not be presented as full bidomain simulation or as a direct model of every arrhythmic mechanism.

EP-PINNs used an Aliev–Panfilov-type reaction–diffusion framework to infer electrophysiological properties from transmembrane-potential observations, with synthetic and experimental optical-mapping tests [42]. Its model states and inference targets differ from activation-time reconstruction. This distinction explains why successful performance on one task cannot be transferred automatically to the other.

The present review identifies model hierarchy as a key comparison axis. Eikonal, monodomain, and bidomain models retain different information [16]. A faster reduced model may be appropriate for an activation-map task but inadequate for evaluating an intervention that depends on detailed cellular dynamics. Tests should match the physiological quantity and disease mechanism that motivate the application.

A recent operator-learning study uses FNO and kernel operator learning to predict cardiac activation and repolarization times from applied stimulation [115]. Evaluation on synthetic domains and a realistic ventricular geometry demonstrates numerical surrogacy for these timing outputs, rather than independent patient-level validation.

5.3 Biomechanics and Constitutive Learning

Inverse elasticity links measured displacement or strain to material parameters through equilibrium and constitutive equations. Kamali et al. examined spatial recovery of elastic modulus and Poisson's ratio using PINNs, including synthetic examples and experimental validation [49]. The study's two-dimensional linear-elastic setting is important when interpreting its applicability to nonlinear, anisotropic, three-dimensional tissue.

Liu et al. learned a constitutive representation from biaxial aortic tissue data and compared it with established material models [60]. This differs from inferring a displacement field by minimizing a PDE residual: the principal learned target is the material response. The surrounding continuum-mechanics assumptions remain essential [33].

Operator learning addresses repeated simulation across material heterogeneity. Yin et al. used DeepONet with phase-field finite-element data for aortic-wall damage and dissection responses [111]. Goswami et al. trained an operator-based surrogate using finite-element growth-and-remodeling data to identify contributing mechanobiological insults [39]. Both demonstrate mechanistic simulation surrogacy; synthetic training data do not establish prospective rupture-risk prediction.

The studies reviewed in this subsection demonstrate elastic-property inference, constitutive learning, and simulation surrogacy for biological tissues. Their findings are specific to the tested material assumptions, loading conditions, and geometries. Extension to musculoskeletal modelling, rehabilitation devices, or prostheses remains a research opportunity rather than a demonstrated outcome of these studies. Such applications would require appropriate muscle, contact, and device–tissue interaction models, together with independent biomechanical validation.

5.4 Neuroscience and Molecular Transport

Brain transport provides a demanding inverse PDE problem because observations are sparse in time and spatial coefficients are uncertain. Zapf et al. compared PINN-based diffusion estimation from MRI with a finite-element inverse approach [112]. Their results emphasize residual control and adaptation: consistency with the numerical comparison depended on sufficiently small trained residuals. This is a useful counterexample to the idea that adding a physics term is sufficient without careful optimization.

Inferred diffusivity is conditional on the tracer, observation timescale, domain representation, and mapping from image intensity to concentration. A diffusion estimate alone does not establish a complete model of clearance or disease progression.

5.5 Pharmacokinetics and Pharmacodynamics

Compartmental models offer a direct connection between ODE learning and drug concentration trajectories. Lu et al. evaluated a neural ODE on trastuzumab emtansine data using different dosing regimens, alongside machine-learning and pharmacometric comparisons [65]. Cross-regimen prediction was more informative than mixing both regimens into one split. The result concerns the studied data and regimens, rather than establishing universal extrapolation for other drugs.

Pharmacokinetic evaluation should specify dosing events, elimination assumptions, covariates, and the physiological interpretation, if any, of learned states. Accurate temporal concentration prediction does not establish spatial drug-delivery accuracy; that claim requires evidence for tissue transport and exchange.

Giacometti et al. studied dalbavancin population pharmacokinetics using data from 218 patients, cross-validation, and comparisons with compartmental and nonlinear mixed-effects models [38]. Covariate handling and augmented pretraining affect the comparison; these results should not be interpreted as evidence for arbitrary dosing or external populations. Complete split and preprocessing specifications are needed when reproducing the reported prediction advantage.

Endocrine physiology provides a complementary hybrid example. De Rooij et al. coupled a shared learned production term to a two-compartment c-peptide model, with person-specific conditioning parameters [26]. Their 117-person dataset was supplemented by an independent 20-person dataset and a physiological comparator. Shared network parameters were fixed when calibrating validation/test individuals. This is transferred population dynamics with individualized calibration, rather than prediction without observing the new individual; identifiability of personalized parameters must be distinguished from joint identifiability of network weights and physiology. A complementary mechanistic approach combines identifiability-informed parameter subset selection with Bayesian inference in a minimal PBPK brain model [25]. It links estimable parameter selection to uncertainty analysis.

5.6 Oncology and Tumor Growth

Reaction–diffusion models describe tumor-cell spread and proliferation, but routine imaging provides indirect and incomplete information about cell density. Zhang et al. estimated patient-specific glioblastoma model parameters using a PINN framework and structural MRI, with synthetic and patient-data evaluations [113]. The work illustrates the importance of nondimensional parameters and complex anatomical domains.

A single imaging snapshot may not distinguish diffusivity, proliferation rate, disease onset, and observation-model assumptions. Later observations can assess predicted progression, while a claim of treatment benefit requires separate clinical evaluation. Learned tumor-model corrections should also be examined for sensitivity to treatment effects, heterogeneity, and omitted mechanics before they are assigned a biological interpretation.

A useful caution comes from Blaom and Okon's preprint comparison of mechanistic and neural ODE tumor-growth models on clinical lesion trajectories [8]. Future observations were held out; the generalized Bertalanffy model performed favorably on average, and added neural flexibility did not establish a universal predictive advantage. This counterexample motivates comparison with parsimonious mechanistic models and reporting calibration failures, rather than selecting a neural model solely for its expressive capacity.

5.7 Respiratory Systems and Pulmonary Modeling

Respiratory simulations couple airflow with geometry, breathing conditions, and sometimes particle transport. Talaat et al. investigated PINNs across a duct, a simplified mouth–lung model, and a patient-specific upper airway [96]. This is simulation and reconstruction evidence in anatomical flow domains; the use of a patient-specific geometry does not establish a clinical diagnostic benefit.

Validation should target the intended airway output, such as pressure drop, resistance, or particle deposition. Evidence obtained under steady incompressible assumptions does not establish accuracy for turbulent flow, moving walls, or multiphase transport.

5.8 Cell Migration and Prospects for Tissue Engineering

Lagergren et al. used biologically informed networks to learn diffusion and growth responses from cell migration assays within a reaction–diffusion framework [56]. The experiments provide evidence beyond synthetic PDE examples and illustrate how neural terms can guide selection of more interpretable mechanistic forms.

Extending cell-assay results to whole-tissue repair requires additional evidence concerning scaffold properties, transport, mechanics, and cell interactions, together with tests across assays and initial cell densities.

5.9 Medical Imaging and Image-Based Modeling

Medical imaging supplies geometry and indirect physiological observations for many inverse models. Van Herten et al. used PINNs for myocardial perfusion MRI quantification, combining kinetic constraints with image-derived time courses [101]. This application is an inference problem within an imaging pipeline, rather than generic image classification or replacement of the MRI acquisition process. Physics-informed perfusion estimation also extends to cerebral imaging. Rotkopf et al. recover tissue response functions from dynamic susceptibility contrast MRI, with simulation-based evaluation and comparisons on glioma patient imaging [84]. Galazis et al. estimate cerebral hemodynamic parameters from infant arterial spin labeling MRI using a spatial uncertainty-based PINN [34]. Their acquisition models and evaluation protocols differ.

Other imaging applications address different physical targets. Configuration-state MRI modeling has been used to investigate neural estimation of relaxation parameters and flip angle [1]. A physics-informed registration framework couples image alignment with large-deformation elasticity and growth or atrophy dynamics [4]. Both retain a physical imaging or deformation model.

Segmentation, registration, motion, sampling, and the image-to-state mapping introduce uncertainties that propagate into inferred coefficients. Quantitative outputs should be checked independently, and sensitivity to plausible preprocessing changes should be reported.

5.10 Patient-Specific Models and Digital Twins

Individualized modeling requires a link between patient observations and model parameters, while a digital twin additionally implies a defined coupling or updating process [18]. SciML can assist calibration, reduce repeated-solve costs, or approximate components of the model. Its role should be stated precisely: a surrogate, an inverse estimator, a data-assimilation component, or an integrated system.

Recent cardiovascular perspectives discuss the principles, potential clinical roles, and AI-enabled development of digital twins [92, 97]. A complete twin additionally needs patient linkage, updating, and intended-use evaluation.

Patient and acquisition-setting transfer require intended-use evidence [104]; anatomy and inference speed alone do not establish deployment readiness.

Salvador et al. learned four-chamber pressure–volume dynamics using 405 electromechanical simulations from one patient-informed anatomy: 400 for training/validation and five held-out numerical tests [89]. The authors report relative errors of approximately 2–6% and assess chamber pressure–volume time traces using normalized root mean square error (NRMSE) and R^2. This demonstrates numerical surrogate fidelity and efficient sensitivity/inference workflows, but five simulation cases are not five patients, and patient-informed geometry is not independent clinical outcome validation.

Figure 4 maps representative biomedical models to learning tasks. It is a conceptual guide, not a claim that every application has the same evidence strength.

Table 4 distinguishes simulation, measured-data evaluation, and clinical effectiveness. Table 5 adds study-specific designs and findings for selected recent comparisons.

Table 5 shows why method selection should follow the computational task. For repeated predictions within a defined simulation family, neural operators and learned reduced dynamics are candidates, provided their offline costs and surrogate errors are assessed. For partially known mechanisms, UDEs offer a way to learn missing terms, but personalized calibration and parameter identifiability require separate evaluation. For temporal forecasting, neural ODEs should be compared with parsimonious mechanistic models using future-time holdouts. The reviewed studies therefore support task-specific comparisons rather than a universal ranking of neural methods.

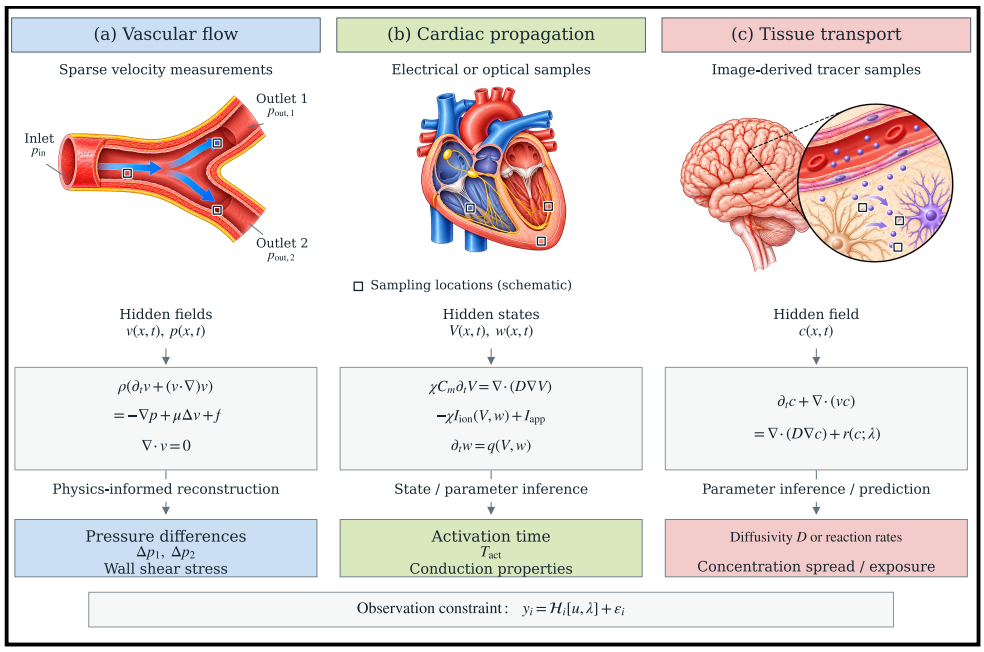


Figure 4: Schematic biomedical applications of physics-informed learning: (a) vascular flow reconstruction; (b) cardiac state and parameter inference; and (c) tissue transport inference and prediction. Governing equations and a shared observation model connect measurements to hidden states and parameters. Colored boxes indicate target quantities and parameters, rather than reported results.

6 Comparative Analysis

6.1 Forward Solution, Inverse Inference, and Repeated Queries

The first comparison criterion is the computational question. For one well-posed forward problem with known coefficients and boundaries, a mature numerical method is often a strong baseline. For an inverse problem, both numerical optimization and a PINN combine an equation with observations, although they implement constraints and sensitivities differently. Zapf et al. provide a direct biomedical illustration [112].

Operator learning targets many queries from a specified problem family. A trained surrogate may evaluate them rapidly, but its training set and offline construction must be included in the comparison. Simulation-based aortic operators illustrate the relationship between a high-fidelity teacher and a learned surrogate [39]. A speedup in this regime does not imply that operator training is efficient for a single new problem.

A surrogate can accelerate an inverse optimizer, and a PINN can learn coefficient-conditioned solutions. In such cases the primary role, amortization assumptions, and retraining requirements should be stated. Method labels alone are insufficient to explain performance.

6.2 Accuracy and Physical Consistency

A useful reporting framework distinguishes representation, discretization, optimization, data error, and model discrepancy. These contributions interact; this framework is not an additive error theorem. Comparison should evaluate the physical quantity used in the biomedical task, not only the state variable easiest to predict.

Residual accuracy differs from agreement with observations. A field can satisfy an incomplete equation while disagreeing with physiology, or fit noisy measurements while violating conservation between them. Evaluation should include independent data and physics quantities. PINN failure analyses reinforce the need to inspect the trained solution rather than infer accuracy from a physics penalty [55].

For transport, an informative integral balance is

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega c\,\mathrm{d}x + \int_{\partial\Omega}\mathbf{J}\cdot\mathbf{n}\,\mathrm{d}s - \int_\Omega r\,\mathrm{d}x = 0. \tag{18}$$

Table 4: Selected original biomedical studies: computational role, evaluation source, output, and evidence limit. This illustrative map is neither exhaustive nor a performance ranking.

Study	Model/role	Evaluation source	Output	Evidence limit
Kissas [53]	Arterial inverse PINN	Flow MRI; model tests	Pressure inference	Reduced model; observation assumptions
Cai [12]	Flow reconstruction	Microaneurysm chip	3D velocity	In vitro; no clinical outcomes
Arzani [5]	Inverse flow PINN	Sparse data; numerical reference	Near-wall flow	Boundary-gradient sensitivity
Villié [105]	Flow-MRI PINN	In vitro stenosis MRI	Artifact correction; pressure/stresses	Bench evidence
Pegolotti [75]	Graph ROM	Cardiovascular simulations	Reduced dynamics	Simulation-surrogate scope
Sahli Costabal [88]	Eikonal PINN	Simulated activation	Activation mapping	No full action potentials
Herrero Martin [42]	Aliev–Panfilov PINN	Synthetic; optical mapping	Parameter inference	Reduced physiology
Cardiac operators [115]	FNO/KOL	Synthetic; ventricular geometry	Activation/repolarization times	Numerical evaluation
Kamali [49]	Inverse elasticity	Synthetic; experimental	Elastic properties	2D linear elasticity
Liu [60]	Constitutive learning	Biaxial aortic measurements	Material response	Tested loading range
Yin [111]	Damage DeepONet	Phase-field FEM	Damage surrogacy	No clinical-risk validation
Goswami [39]	Remodeling DeepONet	Synthetic FEM	Insult identification	Simulated mechanobiology
Van Herten [101]	Perfusion PINN	Synthetic; patient MRI	Perfusion parameters	Kinetic/imaging assumptions
Rotkopf [84]	DSC perfusion PINN	Simulation; glioma MRI	Tissue response/perfusion	Comparator agreement
Galazis [34]	ASL perfusion PINN	Infant MRI	Cerebral hemodynamics	Small imaging cohort
Lu [65]	PK Neural ODE	Clinical cross-regimen tests	Later-cycle prediction	Early patient inputs required
Zhang [113]	Tumor inverse PINN	Synthetic; patient MRI	Infiltration inference	Inverse ambiguity
Zapf [112]	Diffusion PINN/FEM	Synthetic; brain MRI	Tracer diffusivity	Temporal overfitting risk
Lagergren [56]	Reaction–diffusion learning	Cell-migration assays	Term discovery	No whole-tissue validation
Talaat [96]	Airflow PINN	Anatomical simulations	Airway-flow fields	No clinical decisions

For Equation (6), $\mathbf{J} = \mathbf{v}c - \mathbf{D}\nabla c$, with $\mathbf{n}$ the outward unit normal; the balance assumes a fixed domain. Evaluating this relation over independent regions can expose errors obscured by average metrics. Exact discrete balance and accurate local physiology remain separate properties.

Derived quantities such as pressure differences, wall stress, conduction velocity, and concentration exposure require dedicated checks. Differentiation and nonlinear transformations can amplify small field errors. Acceptance criteria should consequently be specified before selecting the network or loss.

6.3 Cost and Scalability

Total cost for N queries can be organized as

$$T_{\mathrm{AI}} = T_{\mathrm{data}} + T_{\mathrm{train}} + T_{\mathrm{tune}} + NT_{\mathrm{infer}}. \tag{19}$$

Against a baseline costing NT_{solve}, the schematic break-even condition is

$$N > \frac{T_{\mathrm{data}} + T_{\mathrm{train}} + T_{\mathrm{tune}}}{T_{\mathrm{solve}} - T_{\mathrm{infer}}}, \tag{20}$$

when $T_{\mathrm{solve}} > T_{\mathrm{infer}}$ and accuracy and hardware are comparable. This is an accounting relation proposed for reporting, not a new benchmark result. Shared numerical setup, reduction cost, retraining, and preprocessing should be included where relevant.

Training costs include failed runs and material hyperparameter searches. GPU inference should not be compared uncritically with an untuned CPU solver. Precision, batching, tolerances, output resolution, and reference convergence all affect results. Memory and sensitivity computation may dominate three-dimensional inverse modeling.

Table 5: Selected 2024–2026 studies: evaluation designs, comparators or references, reported findings, and interpretation limits. Sample sizes and results are study-specific and are not pooled to rank methods. A comparator is an alternative method; a reference is the numerical solution or measurement used to assess predictions.

Study and learned role	Evaluation design and comparator/reference	Reported finding	Limits of interpretation
De Rooij et al. (2025) [26]; conditional UDE	117 individuals and an additional independent 20-person dataset; physiological-model comparator; new individuals undergo personalized recalibration	C-peptide production modeling with individualized conditioning	Evaluation includes individual recalibration; it does not establish prediction without new-individual observations
Giacometti et al. (2025) [38]; Neural ODE for pharmacokinetics	218 patients; six-fold cross-validation; comparison with compartmental and nonlinear mixed-effects models (NLME)	Without covariates, the adjusted KS test did not detect a difference between NODE and NLME residual distributions ($p = 0.51$); improvement depended on covariate configuration	A nonsignificant difference does not establish equivalence; no external-cohort or cross-regimen test in the reported evaluation
Velikorodny et al. (2025) [102]; multifidelity DeepONet	68 held-out bench cases; reference: experimental pressure measurements	Reported experimental–model instantaneous-pressure differences of 1.2 ± 3.5 mmHg	In vitro evaluation with idealized geometries; the reported statistic does not establish clinical-outcome performance
Ryan et al. (2026) [87]; PINO with Bayesian inference	Synthetic inverse tests and data from two Fontan patients; flow MRI for calibration and cuff-pressure measurements as a pressure reference	Flow calibration, pressure consistency, and posterior-uncertainty estimation	Two patients; population transfer and unique recovery of vascular parameters are not established
Salvador et al. (2024) [89]; latent Neural ODE	One patient-informed anatomy; 400 training/validation simulations and five held-out test simulations; reference: electromechanical simulations	Pressure–volume trajectory approximation; chamber-specific time traces assessed using NRMSE and R^2	One anatomy and numerical references; held-out simulations are not independent patients
Csala et al. (2025) [19]; constrained neural dynamics	Idealized simulations of stenosed-vessel flow; comparator: a one-dimensional finite-element model	Improved reduced-model predictions relative to the 1D FEM baseline	Simulation-based evaluation; transfer to new patient geometries remains untested
Blaom and Okon (2025) [8]; neural and mechanistic ODE comparison	Clinical lesion time series with future-time holdouts; mechanistic comparators include the generalized Bertalanffy model	A parsimonious mechanistic model was competitive on average	Preprint evidence; findings depend on the data regime and calibration failures

Mesh size, trajectory length, and the number of anatomies are distinct scaling axes. Increasing network parameter count is not an adequate proxy for all of them. A credible report should identify which axis is being accelerated and whether the accuracy target remains fixed.

6.4 Generalization and Robustness

Interpolation over a simulated parameter range, prediction on a new anatomy, and transfer to a new clinical center are distinct tests. A random split of snapshots from one subject can overestimate patient-level performance. Entire subjects, simulations, or geometries should be held out when the claim concerns those units.

Denser interpolation is not temporal forecasting. A coordinate network can produce a smooth field at additional times without accurately predicting later physiology. PK cross-regimen evaluation illustrates why a changed intervention is more informative than a same-distribution split [65]. Boundary- and intervention-shift tests should match the intended deployment.

Robustness should cover observation noise, missingness, coefficient uncertainty, segmentation perturbations, and model misspecification. Perturbations must be physiologically meaningful. Artificial Gaussian noise alone does not represent every acquisition or preprocessing error.

Table 6: Task-oriented selection guide proposed by this review. Candidates require task-specific comparison; cited studies illustrate relevant evaluation settings rather than establish universal method superiority.

Task and available information	Candidate approaches	Main selection caveat	Decisive evaluation and illustrative evidence
Single forward solve with known equations	Established numerical solver	A learning-based replacement requires justification for the specified task and computational budget	Mesh/time-step convergence and errors in quantities of interest at matched total cost
Sparse observations with unknown coefficients	PINN or numerical inverse optimization	Sparse data may leave parameters non-identifiable under either approach	Noise sensitivity, independent observations, parameter uncertainty, and comparison with numerical inverse estimation [112]
Known mechanistic structure with an unknown response or constitutive component	UDE or learned constitutive law	Learned terms may compensate for errors in the mechanistic model or observation process	Physical admissibility, physical–neural confounding, and prediction under new conditions [26, 60]
Repeated field queries over a related problem family	DeepONet, FNO, graph operator, or ROM	Fast evaluation must offset data-generation and training costs; transfer requires separate testing	Total offline/online cost, held-out instances, and geometry/resolution transfer [39, 111]
Irregular temporal observations with known inputs	Mechanistic ODE, Neural ODE, latent ODE, or controlled ODE	Interpolation accuracy does not establish prediction for new individuals or input regimes	Individual-level splits, future-time prediction, new dosing/input conditions, and solver sensitivity [38, 65]
Unresolved component of a coarse numerical solver	Learned flux, closure, or numerical correction	A locally accurate correction may destabilize the coupled solver	Discrete conservation, consistency, and long-horizon stability against the uncorrected solver [6]
Mechanism discovery from observed fields	Neural PDE or sparse equation discovery	Different candidate mechanisms may explain the same noisy or partially observed data	Recovery under independent conditions, noise sensitivity, and comparison with alternative mechanisms [56, 63]

6.5 Identifiability, Interpretability, and Uncertainty

An interpretable coefficient or learned correction need not identify a unique mechanism. Uncertainty includes measurement, training, numerical, and structural sources; Bayesian PINN estimates remain conditional on the assumed model [110]. Assess calibration and coverage under changed conditions, including uncertainty in clinically relevant thresholds.

Predictive accuracy and parameter identification require different evidence. Neural ODE forecasting does not establish mechanism [8]; conditional UDE transfer may require personalized calibration [26]; and operator-assisted parameter inference remains an inverse problem [87]. InVAErt and qualitative-observation studies illustrate non-identifiable parameter sets and observation-dependent uncertainty [29, 98]. Accordingly, test parameter trade-offs for inverse methods, future-time and disjoint-subject prediction for dynamics, and surrogate error and held-out instances for operator-assisted inference. Experimental operator evaluation provides complementary measured-data evidence [102].

6.6 Task-Oriented Method Selection

Table 6 connects the available information and intended task to candidate methods and decisive checks. Section 6.7 develops three examples showing how the evidence changes the selection question.

Selection requires a task-matched baseline, a computational budget, and evaluation under intended-use conditions. Inverse inference needs identifiability assessment; repeated queries need offline cost accounting; temporal prediction needs individual and input-regime transfer tests.

6.7 Evidence-Based Method Selection

The following comparisons connect the criteria in Table 6 to study-specific results, evaluation designs, and limits of inference.

6.7.1 Brain Transport: PINNs Versus Finite-Element Inverse Estimation

Zapf et al. estimated a spatially constant scalar tracer diffusivity from brain MRI at 0, 7, 24, and 46 h, alongside synthetic tests [112]. With suitable regularization or training settings, both FEM and PINNs recovered the coefficient

within approximately 10% in noisy synthetic tests. Low PDE weights could instead drive the PINN estimate to its imposed lower bound. For MRI data, one ℓ^1-residual formulation yielded $D = 0.75\ \mathrm{mm^2\,h^{-1}}$, close to the FEM estimate of $0.72\ \mathrm{mm^2\,h^{-1}}$, yet still overfit observations at measurement times. Adaptive residual sampling reduced this behavior; residual-based refinement increased computation time by approximately 25% in that setting. The comparison therefore supports checking residuals throughout space–time, boundary treatment, and training cost in addition to coefficient agreement. It does not establish clinical parameter ground truth or a universal residual threshold.

6.7.2 Pharmacokinetics: Neural ODEs Versus Compartmental Models

Lu et al. studied trastuzumab emtansine in 675 individuals: 647 on a three-weekly regimen (Q3W) and 28 on a weekly regimen (Q1W) [65]. Their cross-regimen evaluation used Q3W individuals for model development and Q1W individuals for testing; first-cycle PK observations were available as predictive inputs. For this direction of transfer, the Neural ODE achieved $R^2 = 0.76$, compared with 0.68 for NLME, 0.27 for LightGBM, and 0.15 for LSTM. This supports subsequent-cycle prediction under the tested regimen shift, conditional on early patient observations, rather than prediction without individual measurements or unrestricted extrapolation to other drugs.

Giacometti et al. compared NODEs, two-compartment models, and NLME using 218 dalbavancin patients and six-fold cross-validation [38]. Without covariates, the adjusted Kolmogorov–Smirnov test did not detect a difference between NODE and NLME relative-residual distributions ($p = 0.51$); this is not an equivalence test. With covariates, the authors reported improved NODE predictions and a distributional difference from NLME ($p = 0.008$). The latter comparator used no covariates, its preferred configuration in the study. Thus, the findings concern the complete modeling configurations, including covariate treatment and augmented pretraining. Method selection should test the required patient or dosing shift; cross-validation alone does not establish external-cohort or new-regimen performance.

6.7.3 Operator Surrogates Versus Numerical and Reduced-Order Models

Yin et al. used phase-field FEM simulations for aortic dissection, with 1900 training and 200 test cases in the reported pressure–volume comparison [111]. Test relative errors were 3.70% when true damage fields were supplied and 4.88% when predicted damage fields were used, compared with 8.62% for DeepONet without damage inputs and 6.67% for linear regression. These are pressure–volume response errors, not damage-field errors. The difference between true and predicted damage inputs exposes the effect of upstream surrogate error and input information on downstream accuracy. Goswami et al. likewise used simulated growth-and-remodeling responses for insult identification, comparing sparse-sensor and full-field representations [39].

These examples support simulation-family approximation, not superiority over a matched ROM or clinical risk prediction. An operator is a candidate for repeated queries when its validated output accuracy and query volume justify its construction cost. Equation (20) provides the accounting criterion; the cited accuracy results do not determine a numerical break-even query count.

7 Datasets, Benchmarks, and Reproducibility

7.1 Data Types and Public Resources

Simulation data provide complete fields and known parameters but inherit teacher-model and discretization error, as in aortic surrogacy [111]. Experiments such as cell-migration assays [56] provide independent biological observations; preparation, processing, and measurement uncertainty define their scope. Mixed-source studies must identify which analysis uses each source.

MIMIC-IV provides credentialed clinical records [48], not complete PDE reference fields. Cohort selection, time alignment, missingness, patient-level splits, and the observation model require documentation. PDEBench supplies generic simulated PDE families and baselines [95]; performance there does not establish clinical validity.

Biomedical benchmarks need physiological domains, boundary conditions, parameter distributions, acquisition models, converged reference solvers, and appropriately held-out instances. Reference error must be below the differences being tested. Reuse requires verified data/code access and complete preprocessing and numerical settings; publication of a study alone does not establish availability.

Table 7: Reporting checklist proposed by this review for biomedical SciML. Items apply according to the model, data, and intended use; non-applicable items should be identified and briefly justified.

Component	Information to report, where applicable
Intended task and use	Prediction or inference target, intended use, quantities of interest, and evaluation criteria
Governing model	Equations, units, assumptions, known and unknown terms, initial/boundary conditions, and coupling
Geometry and discretization	Geometry provenance, acquisition and segmentation, mesh or sampling representation, quadrature, spatial/temporal resolution, and convergence checks
Observations	Data provenance, sample size and sampling unit, participant or experimental-preparation characteristics, acquisition protocol, observation operator, noise assumptions, and missing-data handling
Data partition and transfer	Training/validation/test definitions, disjoint sampling units, leakage controls, parameter and geometry ranges, and any adaptation or recalibration performed on evaluation cases
Network and objective	Architecture, activations, normalization, learned outputs, constraints, loss terms and weights; collocation sampling or operator-input representation as appropriate
Optimization	Optimizer, learning-rate schedule, stopping criteria, hyperparameter-selection procedure, tuning budget, random seeds, and failed-run costs that materially affect computational comparisons
Solvers and gradients	Numerical integrator, step-size control, tolerances, algebraic solvers, event handling, and sensitivity, adjoint, or differentiation strategy
Baselines and computational cost	Baseline configuration, comparison at matched accuracy or computational budget, hardware, numerical precision, and separate data-generation, training, and inference costs
Verification and validation	Independent residual or convergence checks; field and parameter errors with definitions, units, and normalization; conservation, robustness, uncertainty, and outputs relevant to the intended use
Ethics and permissions	Applicable ethics approval or exemption, consent, data-use permissions, and confidentiality measures
Availability and reproducibility	Versioned code and data, trained models where relevant, preprocessing and configuration files, dependencies, execution instructions, and access conditions or restrictions

7.2 Metrics and Validation Protocols

A field relative error is

$$E_{L^2} = \frac{\|u_{\mathrm{pred}} - u_{\mathrm{ref}}\|_{L^2(\Omega\times[0,T])}}{\|u_{\mathrm{ref}}\|_{L^2(\Omega\times[0,T])}}. \tag{21}$$

Domain, quadrature, and units must be defined. Near-zero reference norms require another scaling or an absolute error. RMSE, maximum error, temporal profiles, and regional errors provide complementary information. A global average can hide a small clinically important region. Parameter errors should be evaluated separately from field reconstruction. Synthetic inverse tests can establish recovery of known coefficients; clinical estimates may lack direct ground truth. Independent physiology, repeatability, or future prediction can then be more informative than a parameter-error number that cannot actually be computed.

Physics metrics include independent residuals, boundary errors, flux balance, positivity violations, and invariant drift. Statistical metrics include calibration and coverage under changed conditions. Efficiency reports should include data generation and training, alongside inference. The appropriate metrics depend on the intended use. Splits must match the claim: disjoint patients, held-out anatomies, specified parameter ranges, or future observations excluded from calibration. Hyperparameter selection should leave the test set independent. Repeated runs should report variation rather than only the best seed.

7.3 Minimum Reproducibility Specification

Table 7 proposes role-dependent reporting requirements, not a claim that every reviewed study meets them. Frameworks such as DeepXDE facilitate implementation [67], but reproducibility also requires preprocessing, data partitions, solver convergence, integrator and gradient settings, hardware, and precision. Restricted patient data need explicit access conditions; synthetic examples should be identified as substitutes rather than original recordings.

For individual-level clinical prediction, TRIPOD+AI supplies complementary reporting guidance [17]; it does not replace numerical verification or documentation of governing equations and discretization.

7.4 Benchmarking Limitations

Heterogeneous equations, observations, outputs, and cost accounting preclude pooled performance rankings across Tables 4 and 5. Shared biomedical tasks should combine realistic anatomical and acquisition variation with mechanistic, numerical inverse, and ROM baselines appropriate to the intended claim.

8 Challenges and Clinical Translation

8.1 Sparse Observations and Model Misspecification

Biomedical inverse problems often contain more uncertain quantities than the observations can distinguish. Boundary conditions, sources, transport coefficients, and learned corrections may compensate for one another. The resulting difficulty is structural: additional network capacity cannot supply missing information. Measurement design and a defensible choice of inference targets are therefore central to translation.

Physiological equations also omit mechanisms. A residual penalty can favor an incorrect model, while a neural correction can improve prediction by absorbing observation bias or missing inputs. Biological interpretation requires evidence that distinguishes these alternatives. The inverse examples in Section 6.7 illustrate why predictive agreement and mechanistic identification must be assessed separately. The relationship between model misspecification and identifiability further cautions against treating an apparently well-constrained parameter estimate as evidence of a correct biological mechanism [10].

8.2 Stiffness, Scale, and Coupled Physics

Fast reactions coupled to slow transport and tissue-scale processes coupled to organ geometry create multiscale dynamics. Numerical integration must resolve the relevant timescales [41]; learned vector fields can introduce additional stiffness. PINNs face a separate optimization-conditioning problem [106]. Stable optimization and stable physical integration are distinct requirements.

An accurate learned component can destabilize an assembled multiphysics model or violate interface conditions. Its operating range must therefore be assessed in the coupled system, including changed loading and time resolution. Agreement in a bulk output can coexist with errors in a local quantity that controls the biomedical decision.

8.3 Uncertainty and Failure Response

The practical challenge is deciding when a prediction is sufficiently supported to use. Residual checks, ensemble disagreement, and distribution-distance measures are candidate warning signals, but their ability to detect changed physiology must be evaluated. Low residuals can coexist with structural model error, and narrow posterior intervals can reflect restrictive assumptions.

An intended-use workflow should connect measurement, parameter, numerical, and structural uncertainty to a defined response: additional observations, reference-solver evaluation, model revision, or deferral of the decision. The consequences of false reassurance and false alarms should inform thresholds. Figure 5 summarizes this assessment and revision cycle.

8.4 Clinical Evaluation and Governance

The translation gap concerns the complete workflow: acquisition, preprocessing, calibration, prediction, and the resulting action. Retrospective reconstruction or simulator agreement can support development, while claims about improved clinical decisions require evaluation in the target population and care setting. Prospective assessment becomes particularly important when predictions guide an intervention. Personalization alone does not resolve this gap.

A systematic review of digital twins in precision cardiology examines implementation barriers alongside technologies and clinical applications [90]. These barriers concern the full clinical workflow.

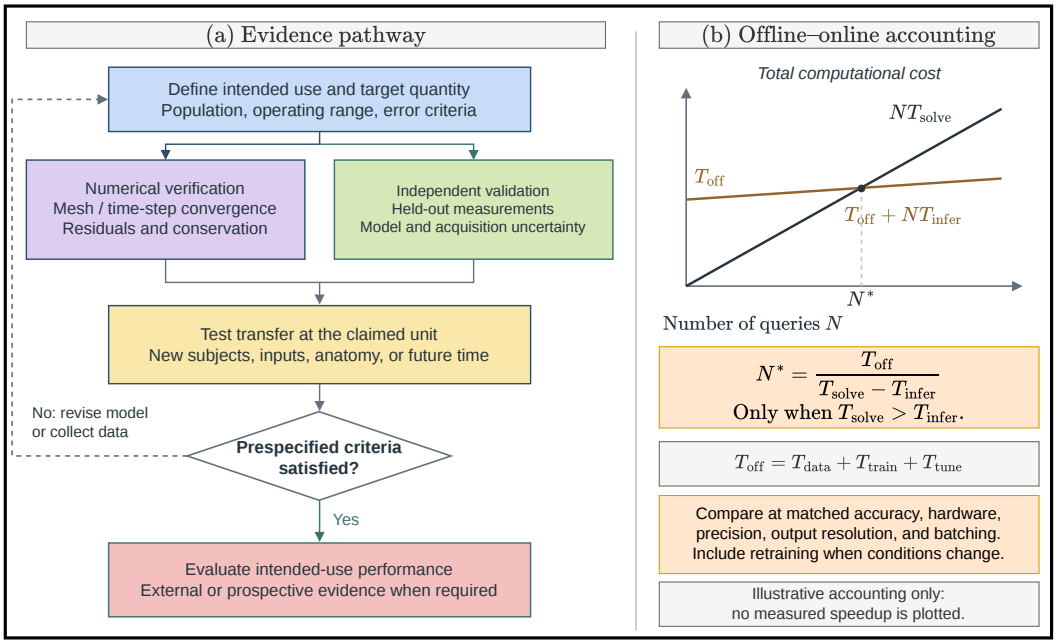


Figure 5: Credibility assessment and computational cost accounting: (a) numerical verification, independent validation, and transfer testing guide model revision and intended-use evaluation; (b) illustrative offline–online cost comparison, with break-even query count N^* when $T_{\text{solve}} > T_{\text{infer}}$. Comparisons require matched accuracy and computational settings. No measured speedup is shown.

FDA guidance for computational modeling and simulation in medical device submissions ties credibility evidence to a defined context of use and model risk [100]. Its scope should be distinguished from general research reporting. For learned components, the relevant considerations include documented operating ranges, provenance, independent evaluation, and change management. Retraining or changing preprocessing may alter performance and require reassessment.

Data governance is also part of the workflow. Publicly accessible papers do not confer permission to reuse patient recordings. This review involved no new participant recruitment or patient-level analysis; future reuse must follow the conditions attached to the original data.

8.5 Limits of This Review

Purposive selection and uneven application coverage constrain this narrative synthesis. It does not estimate method prevalence, establish exhaustive retrieval, or provide a standardized cross-method benchmark. Differences in equations, observations, and evaluation protocols preclude pooled performance rankings. The selection guide, reporting checklist, and credibility pathway are proposed frameworks, not independently validated decision tools.

Preprints are distinguished from published application evidence, and software and literature continue to evolve. These limits particularly affect conclusions about clinical maturity and generalization beyond the selected settings.

9 Future Research Directions

9.1 Output-Specific Error Control

Connect surrogate and discretization errors to decision-relevant pressure, exposure, or stress outputs using a posteriori assessment [103]. Test conservation, positivity, and energy behavior of coupled learned components under changed loading; structure-preserving reduction offers a starting point [21]. Reference-solver fallback should be evaluated including detection and computation overhead.

9.2 Measurement Design for Identifiable Learning

Select observations and perturbations that distinguish competing mechanisms and compensating parameters. Equation discovery [86] and cell-migration modeling [56] motivate learning a flexible response, deriving a compact mechanism, and testing it independently. Biological digital-twin discovery highlights noisy/incomplete data, latent variables, candidate libraries, prior knowledge, and uncertainty as joint challenges [69].

9.3 Transfer and Controlled Updating

Separately test resolution, shape, topology, boundary conditions, and acquisition shifts against matched numerical or ROM baselines [54]. Adaptive twins need explicit updating rules [18], evaluation of delayed or missing observations, and predefined criteria for renewed validation. Report complete-pipeline latency and calibration over successive updates.

9.4 Traceable Pretraining and Shared Benchmarks

Compare PDE pretraining with task-specific learning at matched accuracy, counting adaptation costs and tracing simulated, experimental, and clinical sources. Shared biomedical benchmarks should extend generic resources such as PDEBench [95] with realistic observations and independent measured-data tests. Multi-center evaluation, failure reporting, and an evidence plan tied to context of use [100] are priorities; Table 7 defines the reporting requirements.

Outstanding Questions

- Which measurements distinguish biological parameters from compensating learned terms?
- When do learned solver components retain conservation and stability under changed operating conditions?
- Can operator-assisted inference account for surrogate discrepancy without overstating posterior confidence?
- Which biomedical tasks offset construction costs over realistic numbers of future queries?
- How should evaluation distinguish individual adaptation, temporal forecasting, and transfer to new patients?
- What independent evidence is needed for the specific intended decision?

10 Conclusion

The reviewed evidence supports selecting biomedical SciML methods by computational role and intended use. Sparse-data reconstruction, mechanistic augmentation, temporal prediction, and repeated simulation impose different requirements on training information, numerical integration, identifiability, and validation. A useful comparison therefore connects the learned object to the biomedical quantity of interest and the computation retained at deployment.

Three distinctions are decisive. Accurate field reconstruction does not establish unique parameter recovery; agreement with simulations does not establish clinical validity; and rapid inference must be assessed together with data generation, training, calibration, and retraining costs. These distinctions explain why improvements reported for one task or evaluation setting cannot be transferred automatically to another.

The method-selection guide, credibility pathway, and reporting checklist proposed here translate these findings into a practical assessment framework. Progress toward dependable biomedical SciML requires matched numerical and mechanistic baselines, independent measurements, and explicit tests under changing physiological conditions. The strongest justification for a neural component is a demonstrated improvement in the accuracy, efficiency, or inferential capability of the complete workflow at a specified level of credibility.

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